

Lecture 10: Long-Run Risks

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- Mehra-Prescott leaves too little equity premia unless risk aversion is pushed hard
- Habits and disasters fix this by changing risk prices or tail risk
- Bansal and Yaron (2004) ask a different question: can *small* persistent growth news move prices a lot?
- Long-run growth risk plus time-varying macro uncertainty is their answer
- With Epstein-Zin preferences, distant bad news becomes first-order for valuation

- Motivation: why long-run risk is not “just another $AR(1)$ ”
- A short detour: the extended consumption CAPM
- Primitives: preferences, cash-flow dynamics, and state variables
- Approximate solution via the Campbell-Shiller decomposition ([Campbell and Shiller 1988](#))
- What must be solved numerically in the full model
- Quantitative targets, plus placeholders for the key figures and tables

An Useful Detour

Notation for This Detour

$$m_{t+1} \equiv \log M_{t+1} = \theta \log \delta - \frac{\theta}{\psi} \Delta c_{t+1} - (1 - \theta) r_{w,t+1}$$

$$\tilde{x}_{t+1} \equiv x_{t+1} - E_t[x_{t+1}]$$

- M_{t+1} is the stochastic discount factor
- m_{t+1} is its log
- $r_{w,t+1}$ is the log return on the wealth portfolio
- A tilde means an innovation: the unexpected part relative to time- t information
- We will apply this to $\tilde{m}_{t+1}$, $\tilde{c}_{t+1}$, and $\tilde{r}_{w,t+1}$

Why Start with the Extended Consumption CAPM?

- Recursive utility prices more than current consumption risk
- It also prices shocks that revise continuation values
- Campbell's log-linearization makes those continuation shocks observable as “news”
- This is the clean bridge from Epstein-Zin preferences to long-run risk
- Bansal and Yaron (2004) will load exactly on that extra news term

Two Important Exercises

- Assume: the SDF and the returns of traded assets are jointly lognormal and homokedastic
- Exercise 1: show that, for any asset i , we have

$$\mathbb{E}_t[r_{i,t+1} - r_{f,t}] + \frac{1}{2}\sigma_i^2 = -\text{cov}_t(m_{t+1}, r_{i,t+1}) = \frac{\theta}{\psi}\sigma_{ic} + (1 - \theta)\sigma_{iw}$$

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- Exercise 2: show that the risk-free rate $r_{f,t+1} = \log(R_{f,t+1})$ given by:

$$r_{f,t+1} = -\log(\delta) + \frac{1}{\psi}\mathbb{E}_t[\Delta c_{t+1}] - \frac{\theta}{2\psi^2}\sigma_c^2 - \frac{(1 - \theta)}{2}\sigma_w^2 = \text{constants} + \frac{1}{\psi}\mathbb{E}_t[\Delta c_{t+1}]$$

(Hint: let $i = w$ after exercise 1)

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(Hint: let $i = w$ after exercise 1)

- What's the intuition of ψ ? How does it affect these rates? Why?

$$\tilde{m}_{t+1} = -\frac{\theta}{\psi}\tilde{c}_{t+1} - (1 - \theta)\tilde{r}_{w,t+1}$$

- $\tilde{r}_{w,t+1}$ is the unexpected return on the wealth portfolio
- The wealth portfolio is the claim to aggregate consumption
- Compared with the baseline CCAPM, wealth-return news is also priced
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- Next step: rewrite $\tilde{r}_{w,t+1}$ in terms of consumption news

From Log-Linearization to News Form

$$r_{w,t+1} \approx \kappa_0 + \kappa_1 z_{t+1}^w - z_t^w + \Delta c_{t+1}, \quad z_t^w \equiv \log(P_t/C_t)$$

$$\tilde{r}_{w,t+1} = (E_{t+1} - E_t) [\Delta c_{t+1} + \kappa_1 z_{t+1}^w]$$

- This is the standard Campbell-Shiller log-linearization applied to the wealth claim
- Because the wealth portfolio pays consumption, the dividend-growth term is Δc_{t+1}
- z_t^w is known at time t , so it drops out of the innovation
- Forward-iterate z_{t+1}^w and impose the no-bubble condition
- That produces the news representation on the next slide

$$\tilde{r}_{w,t+1} = (E_{t+1} - E_t) \sum_{j=0}^{\infty} \kappa_1^j \Delta c_{t+1+j} - (E_{t+1} - E_t) \sum_{j=1}^{\infty} \kappa_1^j r_{w,t+1+j}$$

- This is the Campbell-Shiller decomposition in “cash-flow news” and “discount rate news” form
- The dividend on the wealth portfolio is consumption (see Gourio’s notes)
- First term: cash-flow news
- Second term: discount-rate news

A Return Formula in Terms of Consumption News

Let $N_{c,t+1}$ be the news about future consumption growth, so that

$$N_{c,t+1} \equiv (E_{t+1} - E_t) \sum_{j=1}^{\infty} \kappa_1^j \Delta c_{t+1+j}$$

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Exercise 3: show that $(\mathbb{E}_{t+1} - \mathbb{E}_t)[r_{w,t+1+j}] = \frac{1}{\psi}(\mathbb{E}_{t+1} - \mathbb{E}_t)[\Delta c_{t+1+j}]$ using Exs. 1 and 2

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Using exercise 3 we conclude:

$$\tilde{r}_{w,t+1} = \tilde{c}_{t+1} + \left(1 - \frac{1}{\psi}\right) N_{c,t+1}$$

- Intuition for the role of ψ ?

Plugging Back into the SDF

Recall the form of the SDF innovations:

$$\widetilde{m}_{t+1} = -\frac{\theta}{\psi}\tilde{c}_{t+1} - (1 - \theta)\tilde{r}_{w,t+1}, \quad \theta \equiv \frac{1 - \gamma}{1 - 1/\psi}$$

Using what we just derived, we have:

$$\widetilde{m}_{t+1} = -\gamma\tilde{c}_{t+1} - \left(\gamma - \frac{1}{\psi}\right)N_{c,t+1}$$

- **Important:** γ and ψ are not the same parameter!
- You can bump up γ without changing the IES ψ , and vice versa
- What if $\gamma = 1/\psi$?

Questions?

Another Useful Insight

The premium on the wealth portfolio is given by:

$$\mathbb{E}_t[r_{w,t+1} - r_{f,t}] + \frac{1}{2}\sigma_w^2 = \gamma \text{cov}_t(r_{w,t+1}, \tilde{c}_{t+1}) + \left(\gamma - \frac{1}{\psi}\right) \text{cov}_t(r_{w,t+1}, N_{c,t+1})$$

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Don't be lazy:

$$\begin{aligned} \mathbb{E}_t[r_{w,t+1} - r_{f,t}] + \frac{1}{2}\sigma_w^2 &= \gamma \text{cov}_t\left(\tilde{c}_{t+1} + \left(1 - \frac{1}{\psi}\right) N_{c,t+1}, \tilde{c}_{t+1}\right) \\ &\quad + \left(\gamma - \frac{1}{\psi}\right) \text{cov}_t\left(\tilde{c}_{t+1} + \left(1 - \frac{1}{\psi}\right) N_{c,t+1}, N_{c,t+1}\right) \end{aligned}$$

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If innovations in $\tilde{c}_{t+1}$ and $N_{c,t+1}$ are uncorrelated, we get:

$$\mathbb{E}_t[r_{w,t+1} - r_{f,t}] + \frac{1}{2}\sigma_w^2 = \gamma\sigma_{c,t}^2 + \left(\gamma - \frac{1}{\psi}\right) \left(1 - \frac{1}{\psi}\right) \text{var}_t(N_{c,t+1})$$

The Whole Motivation for Bansal and Yaron

Repeating the equation:

$$\mathbb{E}_t[r_{w,t+1} - r_{f,t}] + \frac{1}{2}\sigma_w^2 = \gamma\sigma_{c,t}^2 + \left(\gamma - \frac{1}{\psi}\right) \left(1 - \frac{1}{\psi}\right) \text{var}_t(N_{c,t+1})$$

- Recall that, in the data, σ_c^2 is small
- What happens if consumption growth is i.i.d?
- What if consumption growth moves a lot, but $\gamma = 1/\psi$?
- Bansal and Yaron (2004): what if consumption growth is slowly changing over time and you're afraid of that?
- If $\gamma > 1/\psi$ and $\psi > 1$... we might be back in business!

Primitives for Bansal and Yaron (2004)

- Representative agent with recursive utility ([Epstein and Zin 1989, 1991](#))
- Two aggregate cash-flow objects: consumption and dividends
- The market is a claim to dividends, not a claim to aggregate consumption
- We therefore track both $R_{a,t+1}$ and $R_{m,t+1}$
- Consumption growth will be exogenous and display stochastic volatility

$$\mathbb{E}_t \left[\delta^\theta G_{t+1}^{-\theta/\psi} R_{a,t+1}^{-(1-\theta)} R_{i,t+1} \right] = 1$$

- G_{t+1} is gross consumption growth
- ψ is the IES, γ is risk aversion
- $\theta \equiv \frac{1-\gamma}{1-1/\psi}$
- The whole mechanism needs the Epstein-Zin wedge between γ and ψ

Why Separate $R_{a,t+1}$ and $R_{m,t+1}$?

- $R_{a,t+1}$ is the return on a claim to aggregate consumption
- $R_{m,t+1}$ is the **observable** market return: the aggregate dividend claim
- Solve for the consumption claim first: take $i = a$ in the Euler equation
- Then price the market using (exogenous) dividend dynamics
- There will be two cases: with and without stochastic volatility

Growth Dynamics: Case I (No Stochastic Volatility)

$$x_{t+1} = \rho x_t + \varphi_e \sigma e_{t+1}$$

$$g_{t+1} = \mu + x_t + \sigma \eta_{t+1}$$

$$g_{d,t+1} = \mu_d + \phi x_t + \varphi_d \sigma u_{t+1}$$

- x_t is the small persistent component in expected growth (ρ is close to 1)
- $\phi > 1$ makes dividend cash flows more exposed than consumption
- Notice that σ is constant
- e_{t+1} moves beliefs about future growth, while η_{t+1} is short-run consumption risk
- u_{t+1} adds dividend risk not perfectly tied to consumption
- ρ close to one makes growth news long-lived

Growth Dynamics: Case II (With Stochastic Volatility)

$$x_{t+1} = \rho x_t + \varphi_e \sigma_t e_{t+1}$$

$$g_{t+1} = \mu + x_t + \sigma_t \eta_{t+1}$$

$$g_{d,t+1} = \mu_d + \phi x_t + \varphi_d \sigma_t u_{t+1}$$

$$\sigma_{t+1}^2 = \sigma^2 + \nu_1 (\sigma_t^2 - \sigma^2) + \sigma_w w_{t+1}$$

- Economic uncertainty is the *conditional* variance of consumption growth
- ν_1 controls how persistent volatility shocks are
- w_{t+1} becomes a priced shock in the full model
- This is what generates time-varying risk premia

Questions?

Approximate Solution

Recall: Campbell-Shiller in General

$$R_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$$

$$r_{t+1} \approx \kappa_0 + \kappa_1(p_{t+1} - d_{t+1}) - (p_t - d_t) + \Delta d_{t+1}$$

- This is the standard log-linear return decomposition ([Campbell and Shiller 1988](#))
- P_t : ex-dividend price at time t
- D_t : payout at time t , so next period's cash flow is D_{t+1}
- R_{t+1} : gross return from t to $t + 1$
- Lowercase letters are logs: $r_t = \log R_t$, $p_t = \log P_t$, $d_t = \log D_t$
- It is a first-order expansion around the mean log price-dividend ratio

Campbell-Shiller Step

$$r_{a,t+1} = \kappa_0 + \kappa_1 z_{t+1} - z_t + g_{t+1}$$

- This is the Campbell-Shiller identity applied to the consumption claim
- Set the “dividend” equal to consumption: $D_t = C_t$
- If W_t is cum-dividend wealth, the ex-dividend price is $P_t = W_t - C_t$
- Hence $z_t = \log(P_t/C_t) = \log((W_t - C_t)/C_t)$
- Here Δd_{t+1} becomes consumption growth g_{t+1}
- There is an analogous decomposition for the market using $z_{m,t}$
- This log-linearization makes the economics transparent

Guess the Price-Consumption Ratio

$$z_t = A_0 + A_1 x_t, \quad z_{m,t} = A_{0,m} + A_{1,m} x_t$$

For the consumption claim, take $i = a$ in the Euler equation:

$$1 = \mathbb{E}_t \left[\exp \left(\theta \log \delta - \frac{\theta}{\psi} g_{t+1} + \theta r_{a,t+1} \right) \right]$$

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$$1 = \mathbb{E}_t \left[\exp \left(\theta \log \delta - \frac{\theta}{\psi} g_{t+1} + \theta r_{a,t+1} \right) \right]$$

Conditional normality gives:

$$0 = \mathbb{E}_t[m_{t+1} + r_{a,t+1}] + \frac{1}{2} \text{var}_t(m_{t+1} + r_{a,t+1})$$

- In Case I, the variance term is constant
- Use the guess $r_{a,t+1} = \kappa_0 + \kappa_1(A_0 + A_1 x_{t+1}) - (A_0 + A_1 x_t) + \mu + x_t + \sigma \eta_{t+1}$
- You will only have constants and terms multiplying x_t

Match coefficients to get (you should show this and never trust me)

$$1 - A_1 + \kappa_1 \rho A_1 = \frac{1}{\psi} \implies A_1 = \frac{1 - 1/\psi}{1 - \kappa_1 \rho}$$

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$$1 - A_1 + \kappa_1 \rho A_1 = \frac{1}{\psi} \implies A_1 = \frac{1 - 1/\psi}{1 - \kappa_1 \rho}$$

Redo the analysis for the Market claim in Case I:

$$\phi - A_{1,m} + \kappa_{1,m} \rho A_{1,m} = \frac{1}{\psi} \implies A_{1,m} = \frac{\phi - 1/\psi}{1 - \kappa_{1,m} \rho}$$

- If $\psi > 1$, then $A_1 > 0 \implies x_t$ increases $r_{a,t+1}$
- If $\phi > 1$, dividend claims are more sensitive than consumption claims
- This is the core “small shock, big price effect” step

Exercise 4: show that the following is true:

$$\begin{aligned} m_{t+1} - E_t(m_{t+1}) &= \left[-\frac{\theta}{\psi} + \theta - 1 \right] \sigma \eta_{t+1} - (1 - \theta) \left[\kappa_1 \left(1 - \frac{1}{\psi} \right) \frac{\varphi_e}{1 - \kappa_1 \rho} \right] \sigma e_{t+1} \\ &= \lambda_{m,\eta} \sigma \eta_{t+1} - \lambda_{m,e} \sigma e_{t+1}. \end{aligned} \quad (1)$$

- $\lambda_{m,e}$ is the price of long-run growth news
- As ρ rises, growth news becomes more permanent
- More permanence means more covariance with the IMRS
- The long-run risk premium rises with persistence
- What happens if $\theta = 1$?

Questions?

Case II - Adding Volatility as a State Variable

$$z_t = A_0 + A_1 x_t + A_2 \sigma_t^2$$

- A_1 is unchanged
- A_2 captures how valuation reacts to economic uncertainty
- With $\gamma > 1$ and $\psi > 1$, the paper gets $A_2 < 0$
- Higher uncertainty lowers the wealth-consumption ratio

Time-Varying Premia

In the paper, they show:

$$m_{t+1} - E_t(m_{t+1}) = \lambda_{m,\eta}\sigma_t\eta_{t+1} - \lambda_{m,e}\sigma_te_{t+1} - \lambda_{m,w}\sigma_w w_{t+1}$$

This will imply:

$$\mathbb{E}_t(r_{m,t+1} - r_{f,t}) = \beta_{m,e}\lambda_{m,e}\sigma_t^2 + \beta_{m,w}\lambda_{m,w}\sigma_w^2 - \frac{1}{2}\text{var}_t(r_{m,t+1})$$

where $\text{var}_t(r_{m,t+1}) = \beta_{m,e}^2\sigma_t^2 + \beta_{m,w}^2\sigma_w^2 + \varphi_d^2\sigma_t^2$

- Growth-risk and volatility-risk both earn premia
- When σ_t is high, the conditional risk premium rises
- The model can produce high premia and a low, smooth risk-free rate at the same time
- Said in another way: volatile times are times of low valuation (leverage effect)

What the Approximation Teaches

- Long-run growth news matters because it moves future discount rates and future cash flows
- Volatility news matters because it changes risk premia and current valuations
- The sign restrictions hinge on $\psi > 1$
- Before any numerics, the Campbell-Shiller algebra already tells us what to look for

Questions?

Meeting The Data

What Kind of Equity Risk Premium Can We Get? – Case I

γ	ψ	$E(R_m - R_f)$	$E(R_f)$	$\sigma(R_m)$	$\sigma(R_f)$	$\sigma(p - d)$
Panel A: $\phi = 3.0, \rho = 0.979$						
7.5	0.5	0.55	4.80	13.11	1.17	0.07
7.5	1.5	2.71	1.61	16.21	0.39	0.16
10.0	0.5	1.19	4.89	13.11	1.17	0.07
10.0	1.5	4.20	1.34	16.21	0.39	0.16
Panel B: $\phi = 3.5, \rho = 0.979$						
7.5	0.5	1.11	4.80	14.17	1.17	0.10
7.5	1.5	3.29	1.61	18.23	0.39	0.19
10.0	0.5	2.07	4.89	14.17	1.17	0.10
10.0	1.5	5.10	1.34	18.23	0.39	0.19
Panel C: $\phi = 3.0, \rho = \varphi_e = 0$						
7.5	0.5	-0.74	4.02	12.15	0.00	0.00
7.5	1.5	-0.74	1.93	12.15	0.00	0.00
10.0	0.5	-0.74	3.75	12.15	0.00	0.00
10.0	1.5	-0.74	1.78	12.15	0.00	0.00

What Kind of Equity Risk Premium Can We Get? – Case II

Variable	Data		Model	
	Estimate	SE	$\gamma = 7.5$	$\gamma = 10$
Returns				
$E(r_m - r_f)$	6.33	(2.15)	4.01	6.84
$E(r_f)$	0.86	(0.42)	1.44	0.93
$\sigma(r_m)$	19.42	(3.07)	17.81	18.65
$\sigma(r_f)$	0.97	(0.28)	0.44	0.57
Price Dividend				
$E(\exp(p - d))$	26.56	(2.53)	25.02	19.98
$\sigma(p - d)$	0.29	(0.04)	0.18	0.21
$AC1(p - d)$	0.81	(0.09)	0.80	0.82
$AC2(p - d)$	0.64	(0.15)	0.65	0.67

- Pretty good match for the equity premium and volatility!
- Well-behaved risk-free rate!

Is $\psi > 1$ actually reasonable?

$$g_{t+1} = \mu + x_t + \sigma_t \eta_{t+1}, \quad r_{f,t} = \text{const} + \frac{1}{\psi} x_t + \text{term in } \sigma_t^2$$

- In Hall (1988), the slope of g_{t+1} on $r_{f,t}$ identifies ψ
- That needs $r_{f,t}$ to move only with x_t
- In Bansal and Yaron (2004), $r_{f,t}$ also moves with σ_t^2
- Then $r_{f,t}$ contains extra noise about future growth
- Result: the slope is too small
- In their calibration: true $\psi = 1.5$, estimated $\psi \approx 0.62$ like Hall (1988)

Questions?

$$P_t = \mathbb{E}_t[M_{t+1}X_{t+1}]$$

- This is still the master equation
- The vanilla CCAPM fails in the data; the framework does not
- The big empirical fact is time-varying discount rates
- Good models make marginal utility rise sharply in bad times
- Habit, disasters, and long-run risks are three ways to get there
- Long-run risk does it with small but persistent growth and volatility news

Questions?

Readings and References i

- Campbell, Chapter 6

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